

Exercise 2 – LLM Leaderboard

1. Favorite LLM Rank

Model: GPT-5.6 Sol (max) — OpenAI

Leaderboard: Artificial Analysis Intelligence Index (artificialanalysis.ai/leaderboards/models)

Rank: #5 overall

Score: 61

Other stats: \$1.23 cost per task, 73 tokens/s median speed, ~104s total end-to-end response time

AIML-Lab02/Readme.md

Course: IT2011 - Artificial Intelligence

AIML - Lecture 02

LLM Leaderboard - Comparison

Ask Gemini

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Figure 1: Artificial Analysis leaderboard showing GPT-5.6 Sol (max) ranked #5.

2. New Models Tried

- Qwen3
- Kimi K2.5

3. Prompt 1 (Reasoning) — Bat and Ball Problem

Prompt used: "A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost? Explain your reasoning step by step."

Qwen3 Response

Thinking completed >

The Bat and Ball Problem

The Setup

Let's assign a variable to the unknown:

- Let x = the cost of the ball

Translating the Clues into Math

Clue 1: The bat and ball cost \$1.10 total.

$$\text{ball} + \text{bat} = \$1.10$$

Clue 2: The bat costs \$1.00 more than the ball.

$$\text{bat} = x + \$1.00$$

Solving Step by Step

Step 1: Substitute the bat's price into the total equation:

$$x + (x + \$1.00) = \$1.10$$

Step 2: Combine like terms:

$$2x + \$1.00 = \$1.10$$

Step 3: Subtract \$1.00 from both sides:

$$2x = \$0.10$$

Step 4: Divide by 2:

$$x = \$0.05$$

The Answer

The ball costs **\$0.05** (5 cents).

Verification

Item	Cost
Ball	\$0.05
Bat (\$1.00 more)	\$1.05
Total	\$1.10 ✓

💡 Why do most people get it wrong? The intuitive answer is \$0.10. But if the ball were \$0.10, the bat would be \$1.10, making the total \$1.20 — which doesn't match! This is a famous problem from behavioral economics that demonstrates how our brains often jump to a quick, intuitive answer instead of carefully working through the math.

Figure 2: Qwen3's response to the bat and ball problem.

Kimi K2.5 Response

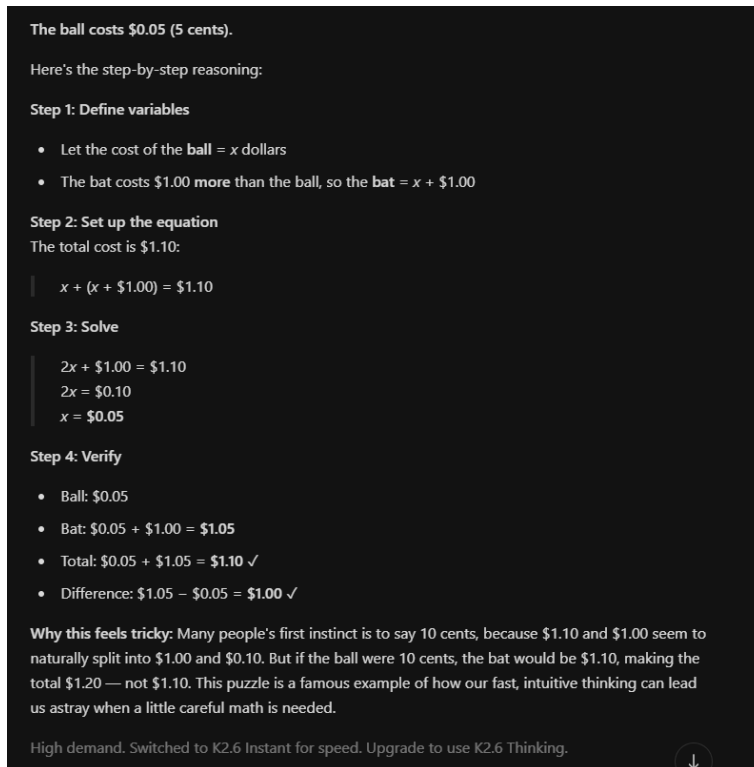


Figure 3: Kimi K2.5's response to the bat and ball problem.

4. Prompt 2 (Practical/Creative) — Study Plan

Prompt used: "Write a short study plan for a university student who has one week to prepare for a probability and statistics exam, assuming they're starting from zero and are also busy with other coursework."

Qwen3 Response

 The 7-Day Survival Plan

Day 1: Probability Foundations

- **Goal:** Understand how to calculate basic and conditional probabilities.
- **Topics:**
 - Basic rules (Addition, Multiplication, Complement).
 - Conditional Probability and Independence.
 - Bayes' Theorem (know how to set up the formula and a tree diagram).
- **Action:** Watch a quick primer on Bayes' Theorem. Solve 5 word problems involving conditional probability.

Day 2: Random Variables & Distributions

- **Goal:** Grasp how probability is modeled for discrete and continuous data.
- **Topics:**
 - Discrete: Probability Mass Function (PMF), Expected Value $E[X]$, Variance $Var(X)$. Focus on Binomial and Poisson distributions.
 - Continuous: Probability Density Function (PDF), Cumulative Distribution Function (CDF). Focus heavily on the Normal (Gaussian) Distribution and standardizing to Z-scores.
- **Action:** Practice converting raw scores to Z-scores and using the standard normal table (or calculator) to find probabilities.

Day 3: The Bridge to Statistics (Sampling & CLT)

- **Goal:** Understand how sample data relates to population parameters.
- **Topics:**
 - Population vs. Sample, Parameters vs. Statistics.
 - Central Limit Theorem (CLT): Know the conditions ($n \geq 30$) and why it allows us to use the Normal distribution for sample means.
 - Standard Error of the Mean ($\sigma / \sqrt{n}$).
- **Action:** Solve 3 problems calculating the probability of a *sample mean* falling within a certain range using the CLT.

Day 4: Confidence Intervals (CI)

- **Goal:** Learn how to estimate population parameters with a margin of error.
- **Topics:**
 - Concept: "We are 95% confident that the true population mean lies between X and Y."
 - Formulas for CI of a Mean (Z-interval if σ known, T-interval if σ unknown) and Proportion.
 - How sample size and confidence level affect the margin of error.
- **Action:** Practice calculating the margin of error and constructing a 95% CI for both means and proportions.

Day 5: Hypothesis Testing (The Highest-Yield Topic)

- **Goal:** Master the logic and mechanics of testing claims.
- **Topics:**
 - Null (H_0) and Alternative (H_a) hypotheses.
 - Type I (α) and Type II (β) errors.

Day 5: Hypothesis Testing (The Highest-Yield Topic)

- **Goal:** Master the logic and mechanics of testing claims.
- **Topics:**
 - Null (H_0) and Alternative (H_a) hypotheses.
 - Type I (α) and Type II (β) errors.
 - The p-value: Know the rule: "If p is low, H_0 must go."
 - Mechanics: 1-sample Z-test and T-test.
- **Action:** Walk through 4 full hypothesis test problems step-by-step: (1) State hypotheses, (2) Check conditions, (3) Calculate test statistic, (4) Find p-value, (5) State conclusion in context.

Day 6: Mock Exam & Targeted Repair

- **Goal:** Simulate exam conditions and identify blind spots.
- **Action:**
 - Take a past exam, a professor-provided practice test, or a comprehensive set of textbook end-of-chapter problems under timed conditions (1.5 hours).
 - Grade it harshly.
 - Spend the remaining 30 minutes reviewing *only* the questions you got wrong. Re-solve them without looking at the solution.

Day 7: Final Review & Formula Consolidation

- **Goal:** Lock in memory and reduce anxiety.
- **Action:**
 - Review your personal "Cheat Sheet" of formulas and concepts.
 - Re-do 2–3 tricky problems from Days 4 and 5.
 - **Stop studying 2–3 hours before bed** to let your brain consolidate the information. Get a full night's sleep.



Figure 4–5: Qwen3's 7-day study plan.

Kimi K2.5 Response

Day 1 — Foundations & Set Theory
• 45 min: Sample spaces, events, basic set operations (union, intersection, complement) • 45 min: Axioms of probability, counting principles (permutations & combinations) • 1 hr: Drill 15–20 basic probability problems (drawing cards, dice, coins)
Day 2 — Conditional Probability & Independence
• 1 hr: Conditional probability, Bayes' theorem, law of total probability • 1 hr: Independence of events, tree diagrams • 30 min: Practice Bayes' theorem problems (medical testing scenarios are classic exam fodder)
Day 3 — Random Variables (Discrete)
• 1 hr: PMFs, CDFs, expectation, variance • 1 hr: Binomial, Poisson, and geometric distributions—know formulas and when to apply them • 30 min: Practice identifying which distribution fits a word problem
Day 4 — Random Variables (Continuous)
• 1 hr: PDFs, CDFs, expectation, variance for continuous variables • 1 hr: Uniform, exponential, and normal distributions • 30 min: Standard normal table practice (Z-scores, finding probabilities)
Day 5 — Joint Distributions & Limit Theorems
• 1 hr: Joint, marginal, conditional distributions; covariance and correlation • 1 hr: Law of large numbers, Central Limit Theorem (CLT)—understand conceptually and know how to apply CLT for approximations • 30 min: Mixed practice problems
Day 6 — Statistics: Estimation & Inference
• 1 hr: Sample mean, sample variance, bias, consistency; method of moments and MLE basics • 1 hr: Confidence intervals (mean, proportion), hypothesis testing (z-tests, t-tests, p-values) • 30 min: Practice writing out full hypothesis test steps (null, alternative, test stat, decision)
Day 7 — Review & Mock Exam
• 1.5 hr: Skim condensed notes/formula sheet; redo problems you got wrong earlier • 1 hr: Timed practice with 1–2 past exams or dense problem sets • 30 min: Sleep early. Cramming formulas at 2 AM has negative returns.

Figure 6: Kimi K2.5's 7-day study plan.